# Multigrid Learned Primal-Dual Splitting for Artifact-Free Accelerated MRI Reconstruction

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## Impact

Compact unrolled networks for accelerated MRI occasionally leave residual aliasing artifacts that undermine trust in their reconstructions. A multigrid extension suppresses these artifacts at matched size and runtime, making algorithm-derived reconstruction networks more reliable.

## Synopsis

**Motivation:** Directly parameterized unrolled networks such as LPDSNet achieve state-of-the-art accelerated MRI reconstruction with few parameters, but can leave residual aliasing artifacts.
**Goal(s):** Suppress these artifacts while keeping LPDSNet's compact, algorithm-derived architecture.
**Approach:** We embed LPDSNet's primal-dual iterations in a multigrid V-cycle derived from MGProx. Coarse grids solve rediscretized versions of the same reconstruction problem, so spatially extended aliasing is corrected cheaply; the network is trained end-to-end.
**Results:** On fastMRI knee and brain at R=8 and 12, MGLPDSNet [removes the aliasing artifacts] of LPDSNet at [matched parameters and runtime], while [matching/exceeding] E2E-VarNet.

## Introduction

Unrolled networks reconstruct accelerated MRI by alternating data consistency with a learned prior. Directly parameterized networks such as CDLNet[1] and LPDSNet[2] go further: every layer is one iteration of a classical algorithm, here primal-dual splitting, so the network is compact, noise-adaptive and interpretable by construction, while matching black-box networks such as E2E-VarNet[3]. In some reconstructions, however, LPDSNet leaves residual aliasing (Figure 3). Undersampling aliasing is spatially global, folding energy across the field of view, whereas each LPDS iteration applies only local filters at full resolution, so a fixed budget of iterations cannot move information far enough to undo it. Multigrid methods remove exactly this kind of spatially extended error by correcting it on coarse grids[4], and MGProx[5] extends them to the nonsmooth problems that LPDSNet solves. We propose MGLPDSNet, which embeds LPDSNet's iterations in a multigrid V-cycle.

## Methods

**LPDSNet:** Given undersampled multicoil k-space $\boldsymbol{y} = \boldsymbol{E}\boldsymbol{x} + \boldsymbol{n}$, with $\boldsymbol{E} = \boldsymbol{M}\boldsymbol{F}\boldsymbol{S}$ (sampling mask, Fourier transform, coil sensitivities), each LPDS layer takes a primal gradient step on the image $\boldsymbol{x}$ and a dual step on latent subband coefficients $\boldsymbol{z}$ (Figure 1): learned analysis and synthesis convolutions $\boldsymbol{A}^H$, $\boldsymbol{B}$ replace the sparsifying transform, and the dual proximal map is a subband-wise clipping $\Pi_{\boldsymbol{\lambda}}$. Step sizes and thresholds are affine in the noise level $\sigma$[1,2].

**MGLPDSNet:** Each iteration is a V-cycle over three grid levels ($N$, $N/2$, $N/4$; Figure 1). LPDS layers at a level are followed by an FAS-Correction, which restricts the state to the next coarser grid and tilts that level's problem so it reproduces the fine level's residual: the full approximation scheme[4], applied to both the primal and the dual residual. Each coarse level solves the same SENSE problem posed on its own grid. Its result returns through a Coarse Correction, prolonging the coarse update with a learned step, before further LPDS layers. MGLPDSNet unrolls [6] V-cycles with [4/4/6] layers per level and $M = [81]$ subbands at every level, all weights untied.

**Experiments:** We trained on multicoil fastMRI[6] knee (proton density) and brain (T2-weighted) data, using measured k-space with added white Gaussian noise of $\sigma \in [0.01, 0.02]$ and 1-D random Cartesian undersampling at R=8 and 12 with a 4% fully sampled centre. Coil sensitivities were estimated with ESPIRiT from the noisy calibration region; targets were SENSE combinations of the fully sampled data. All networks were trained identically (magnitude $\ell_1$+$\ell_2$ loss within the anatomy mask, Adam, learning rate $5 \times 10^{-4}$ with cosine decay, $3 \times 10^5$ steps). Baselines were LPDSNet[2] (30 layers, 169 subbands) and E2E-VarNet[3] given the same coil sensitivities [, and RAM[7] trained under the same protocol]. PSNR, SSIM and NRMSE were computed within the anatomy mask on [N] test

volumes at $\sigma =$ [0.015], with paired Wilcoxon signed-rank tests (Holm-corrected); inference time was measured on [one GPU type].

## Results

Figure 2 shows representative reconstructions for each anatomy and acceleration. [One or two sentences: where MGLPDSNet is visibly better, and what the zoomed regions show.]

Figure 3 shows LPDSNet's residual aliasing: [describe, e.g. a ghost of the anatomy shifted by FOV/R, clearly visible in the error map]. MGLPDSNet reconstructs the same slices without it. [How often it occurs: found in N of M test slices by ...]

Across [N] test volumes (Figure 4), MGLPDSNet [improved on LPDSNet in every metric (p<...)], with [N]× fewer parameters than E2E-VarNet and an inference time of [ms] per slice against [ms]. [Versus E2E-VarNet: ...]

The ablations (Figure 5) show [how quality scales with the number of subbands $M$ and of V-cycles, and the single-resolution LPDSNet row as the no-multigrid reference].

## Discussion

The coarse levels give every V-cycle a path along which information travels across the image in few operations, which is what undoing global aliasing requires; local full-resolution iterations lack it. Because each coarse level solves the same reconstruction problem, the network keeps the structure of its source algorithm rather than becoming a generic multiscale CNN. [Limitations: retrospective undersampling, Cartesian sampling, a single noise model.]

## Conclusion

Embedding LPDSNet in a multigrid V-cycle suppresses its residual aliasing artifacts at [matched size and runtime], while [keeping] state-of-the-art reconstruction quality.

## Acknowledgements

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## References

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# Figures

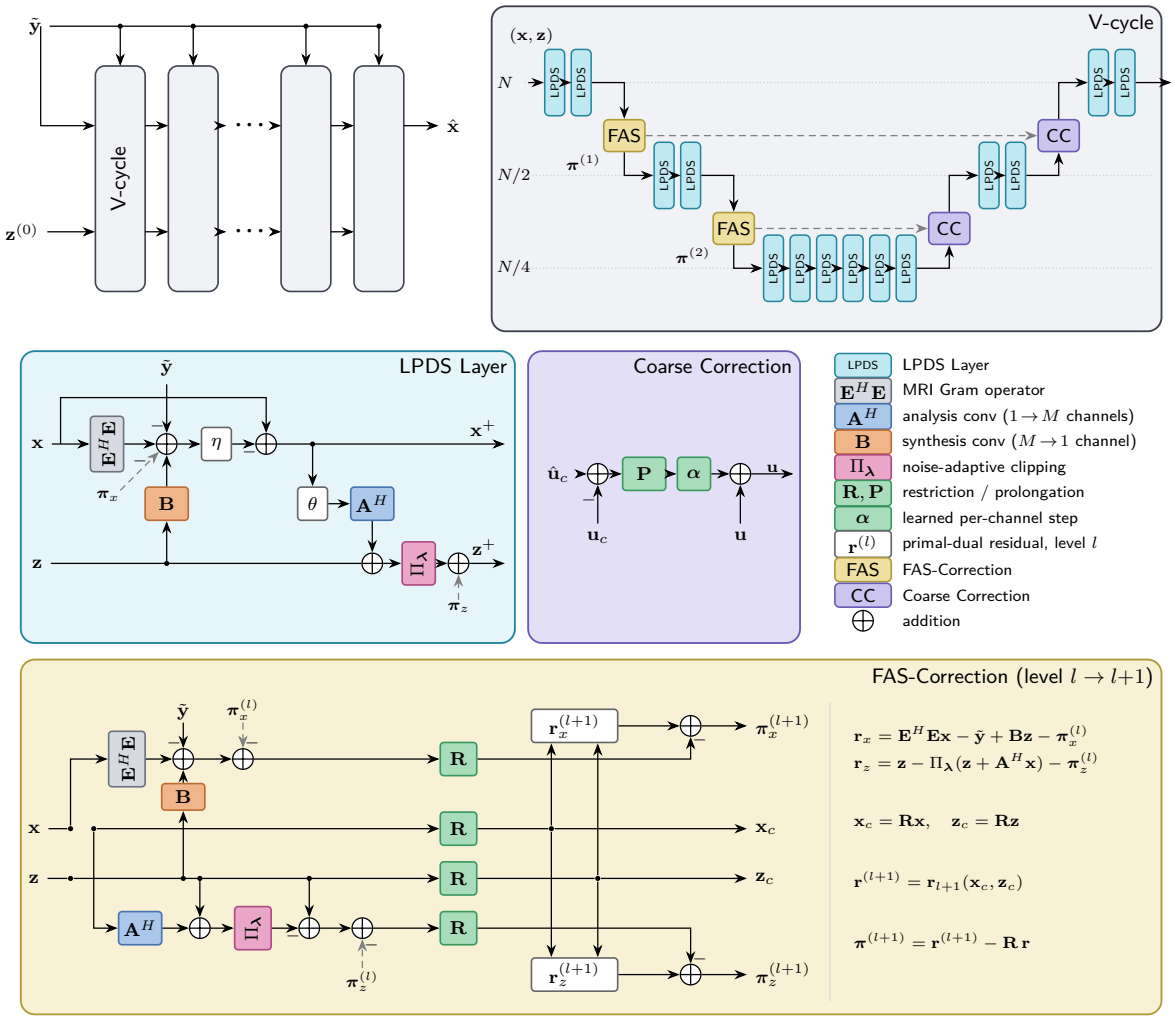



**Figure 1:** MGLPDSNet. Top left: [6] untied V-cycles, each receiving the preprocessed measurement $\tilde{\boldsymbol{y}} = \boldsymbol{E}^H(\boldsymbol{y} - \mu\boldsymbol{E}\mathbf{1})$, $\boldsymbol{E}$ and $\sigma$; the first layer starts from $\boldsymbol{x} = \tilde{\boldsymbol{y}}$, $\boldsymbol{z}^{(0)} = \Pi_{\boldsymbol{\lambda}}(\boldsymbol{A}^H\tilde{\boldsymbol{y}})$. Top right: one V-cycle over three grid levels; each level solves its own problem $(\tilde{\boldsymbol{y}}_l, \boldsymbol{E}_l, \sigma_l)$, tilted by the FAS correction $\boldsymbol{\pi}^{(l)}$. Middle: an LPDS Layer, with $\tilde{\boldsymbol{x}} = \boldsymbol{x}^+ + \theta(\boldsymbol{x}^+ - \boldsymbol{x})$; $\eta, \theta, \boldsymbol{\lambda}$ are affine in $\sigma$; dashed inputs (the FAS terms) exist on coarse levels only. The Coarse Correction acts on $\boldsymbol{u} \in \{\boldsymbol{x}, \boldsymbol{z}\}$ with $\hat{\boldsymbol{u}}_c$ the coarse level's output and $\boldsymbol{u}_c$ its restricted input. Bottom: the FAS-Correction, in the order its operations run; it carries its own learned copies of $\boldsymbol{A}^H$, $\boldsymbol{B}$ and $\boldsymbol{\lambda}$.

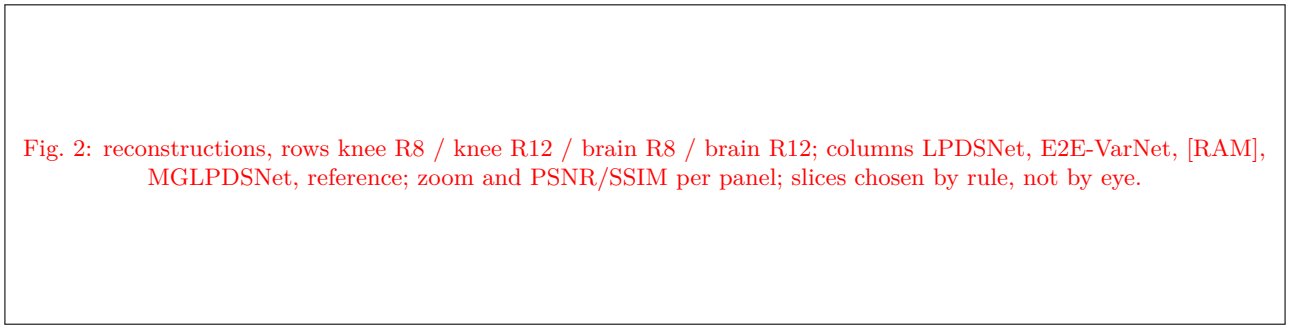



**Figure 2:** [Representative reconstructions ...]

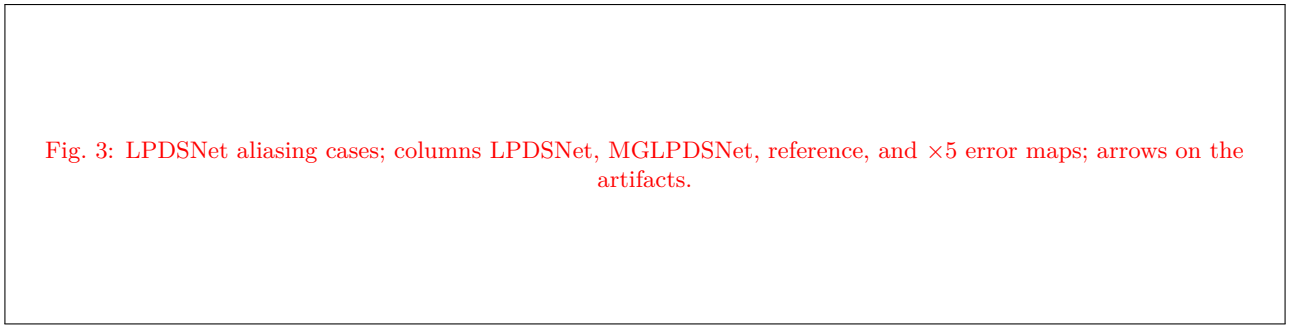



**Figure 3:** [Residual aliasing in LPDSNet ...]

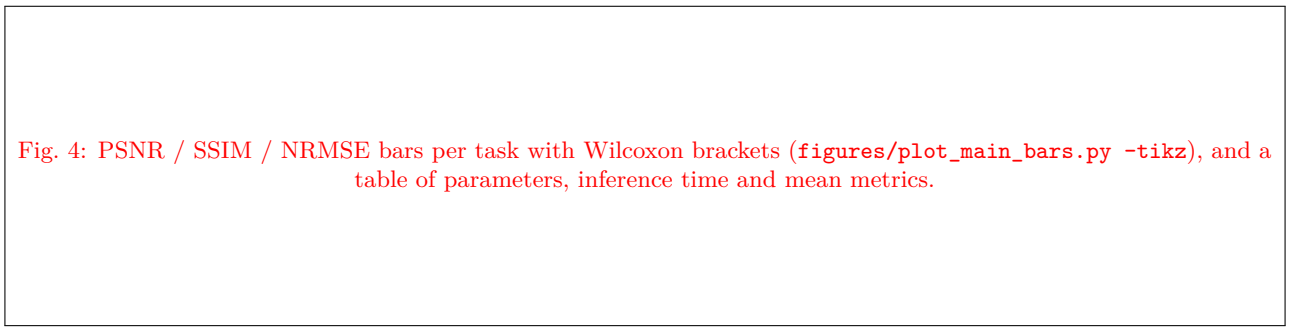



**Figure 4:** [Quantitative comparison ...]

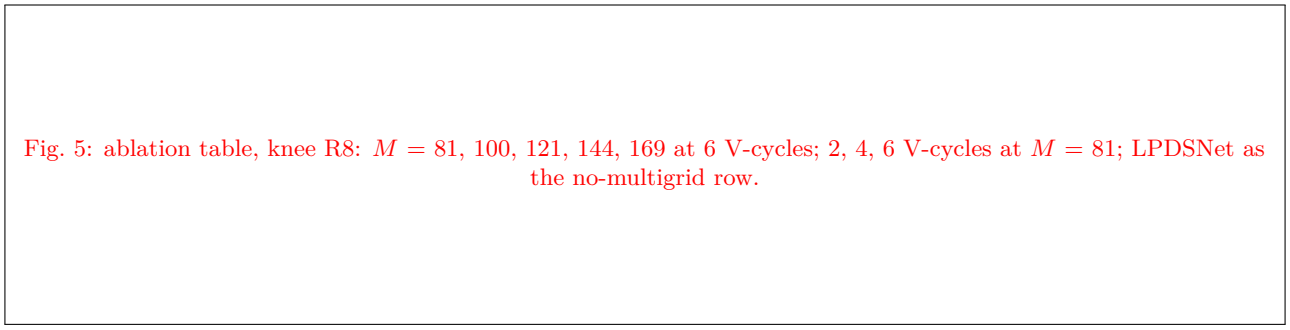



**Figure 5:** [Ablations ...]